

VANNES SENE, France

WMO: 072100

Lat: 47.6045N

Lon: 2.7140W

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.3	30.1	21.6	16.3	31.8	24.8	18.9	33.3	25.9	48.4	24.1	46.8	5.2	40	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	84.0	68.3	79.7	66.3	76.5	64.9	69.5	80.1	67.8	77.4	66.2	73.6	8.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.4	72.5	64.6	91.2	70.6	63.2	86.7	69.1	33.5	80.0	32.1	77.6	30.9	73.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.0	16.9	DB	23.9	89.8	3.0	4.7	21.8	93.2	20.1	96.0	18.4	98.6	16.3	102.1	(4)
(5)			WB	22.7	72.8	3.6	2.6	20.0	74.6	17.9	76.1	15.9	77.6	13.2	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	43.5	43.9	47.0	50.9	56.1	61.9	65.2	63.6	61.2	55.3	48.2	44.8		
	DBStd	9.46	5.89	5.54	5.34	5.31	5.20	5.80	5.09	4.80	5.20	5.30	5.27	5.98		
	HDD50	863	211	179	120	50	8	2	0	0	1	16	96	180		
	HDD65	4353	668	591	557	424	279	125	58	81	136	302	504	628		
	CDD50	2147	9	8	28	76	199	359	470	422	337	180	42	17		
	CDD65	161	0	0	0	0	5	32	64	37	22	1	0	0		
	CDH74	1304	0	0	0	6	45	283	554	268	148	0	0	0		
	CDH80	332	0	0	0	0	4	75	159	64	30	0	0	0		
(14) Wind	WSAvg	8.4	8.5	9.2	9.4	8.9	8.6	8.1	8.3	7.6	7.6	7.9	8.0	8.8		
(15) Precipitation	PrecAvg	35.2	4.3	3.0	2.8	2.8	2.4	1.8	2.1	1.9	2.2	4.2	4.2	4.1		
	PrecMax	44.2	8.3	7.3	7.8	6.1	4.3	4.0	3.6	4.7	6.9	6.5	9.3	8.3		
	PrecMin	23.3	0.9	0.7	0.3	0.8	0.8	0.5	0.4	0.4	0.6	0.7	1.1	1.0		
	PrecStd	4.9	2.2	1.7	1.7	1.4	0.9	1.1	0.9	1.2	1.5	1.7	2.1	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	59.2	65.7	73.9	79.0	89.0	91.1	86.4	84.4	71.2	62.6	57.8		
		MCWB	53.6	51.8	56.0	60.8	65.4	72.5	68.9	68.8	68.1	62.7	58.7	55.2		
	2%	DB	54.8	55.3	60.7	68.5	73.8	80.8	84.2	80.6	78.1	67.6	60.3	56.1		
		MCWB	52.8	51.1	53.0	57.5	63.1	68.0	67.6	66.6	65.0	61.3	57.3	54.2		
	5%	DB	53.6	53.6	57.4	63.8	69.8	76.6	80.2	76.6	73.8	64.9	58.7	55.3		
		MCWB	51.6	50.6	51.3	55.0	60.6	65.3	66.6	64.8	63.3	60.0	56.1	53.4		
	10%	DB	52.2	52.2	55.1	60.1	65.9	73.1	75.9	73.1	70.7	63.2	57.1	53.9		
		MCWB	50.4	49.8	50.9	53.5	58.4	63.1	65.1	63.6	62.0	59.3	54.5	52.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	53.9	56.8	62.2	67.2	73.9	71.3	70.8	69.3	64.5	60.6	56.3		
		MCDB	55.3	56.6	62.3	71.6	76.8	84.9	83.8	81.6	79.6	68.4	61.5	56.8		
	2%	WB	53.4	52.6	54.4	58.7	64.0	68.9	69.2	68.6	66.9	62.7	58.5	55.1		
		MCDB	54.5	54.3	58.1	65.7	72.0	78.7	80.4	77.8	74.7	65.4	59.5	56.0		
	5%	WB	52.2	51.6	53.1	56.2	61.6	66.5	67.5	66.7	65.2	61.5	56.8	53.9		
		MCDB	53.4	52.9	55.9	62.1	68.3	74.4	78.1	73.5	71.5	64.0	58.4	54.9		
	10%	WB	50.9	50.4	51.8	54.3	59.4	64.4	65.8	65.0	63.7	60.3	54.9	52.6		
		MCDB	52.2	51.9	54.2	59.1	64.7	71.4	73.6	70.4	69.0	62.7	56.8	53.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.4	13.6	15.9	16.2	17.8	17.8	17.8	17.5	14.1	12.9	10.9		
		MCDBR	9.8	13.1	17.3	21.9	22.0	23.7	25.5	22.6	21.4	14.7	12.4	9.6		
	5% WB	MCWBR	8.7	10.0	11.2	12.6	12.6	12.2	11.6	10.9	10.5	9.7	9.8	8.7		
		MCWBR	9.1	11.0	13.8	18.7	20.1	21.3	22.2	19.2	18.1	12.3	10.7	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.350	0.378	0.393	0.398	0.397	0.393	0.382	0.367	0.365	0.342	0.329		
		taud	2.411	2.384	2.312	2.285	2.317	2.351	2.366	2.414	2.437	2.433	2.426	2.392		
	Ebn at Noon		233	256	265	272	275	275	274	273	267	247	228	219		
		Edh at Noon	22	28	35	39	39	38	37	34	30	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		336	630	1010	1513	1793	1968	1881	1639	1304	742	425	293		
	RadStd		32	62	102	148	119	178	84	108	90	61	36	32		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page